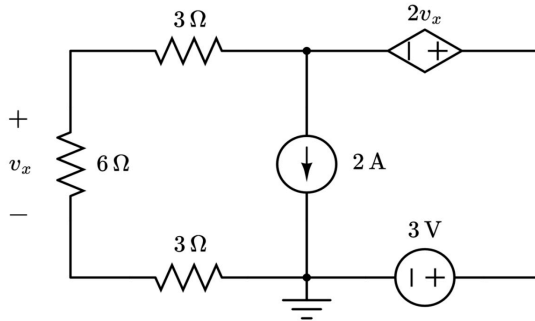


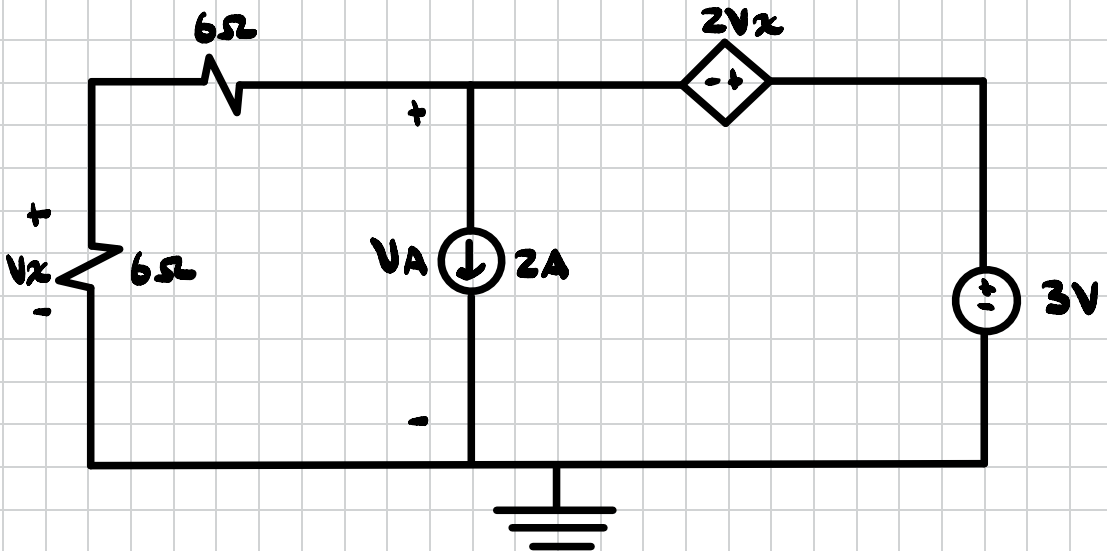
Practice Quiz Four: Node Voltage Analysis (Hard)

For the circuit shown below, use node voltage analysis to find the value of v_x .

Suggested time limit: 15 minutes



Step # 1 Redraw Circuit



Step # 2 Voltage @ V_A

$$-V_A - 2V_x + 3 = 0$$

$$V_A = 3 - 2V_x$$

Step #3 Nodal @ V_x

$$\frac{V_x - 0}{6} + \frac{V_x - V_A}{6} = 0$$

$$\left(\frac{V_x}{6} + \frac{V_x - (3 - 2V_x)}{6} = 0 \right) 6$$

$$V_x + V_x - 3 + 2V_x = 0$$

$$4V_x = 3$$

$$V_x = \frac{3}{4} V$$